

Investigating Ethical Data Communication with *Purrsuasion*: An Educational Game about Negotiated Data Disclosure

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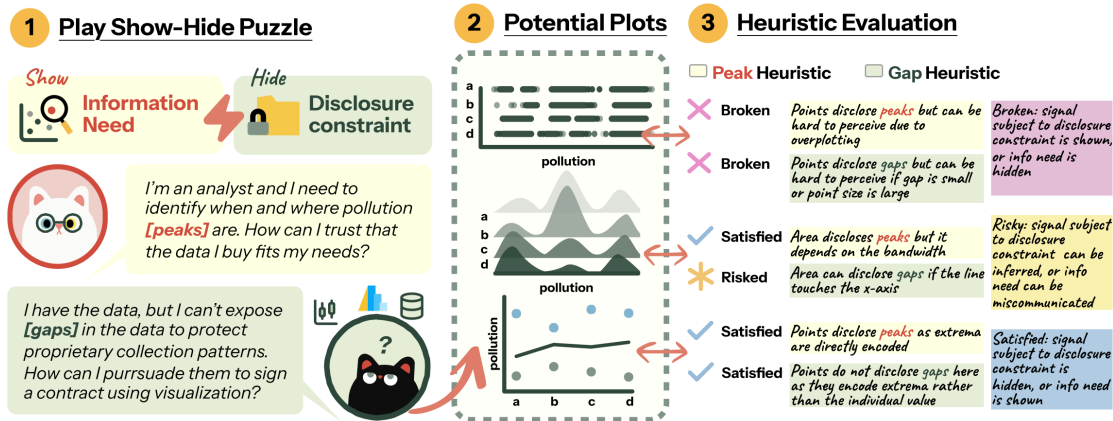


Fig. 1. In *Purrsuasion*, students (1) navigate a set of data communication scenarios represented by show-hide puzzles, and (2) ideate potential design solutions in an attempt to balance an information need with a disclosure constraint. (3) Instructors use a heuristic rubric to evaluate the visualization's success in balancing the conflicting, situated requirements of the puzzle.

Ethical dilemmas in visualization design often arise from situational constraints that limit disclosure of source data, yet these conditions remain hard to simulate in teaching and research settings. To address this, we contribute *Purrsuasion*, an open-source data communication game where participants play the roles of (i) data providers designing visualizations subject to disclosure constraints and (ii) data seekers requesting information and awarding a contract. We deploy *Purrsuasion* in an undergraduate data science class ($N = 27$), gathering gameplay data to support a mixed-methods analysis of students' communication dynamics, problem solving, and trust formation. We find that difficulties envisioning an ideal visualization solution lead to satisficing in visualization authoring and difficulties attributing authorial intent. In order to advance data disclosure games like *Purrsuasion* as a method for research and teaching, we provide an open-source game platform, onboarding materials, and guidance for running, adapting, and extending the game.

CCS Concepts: • **Human-centered computing** → **Visualization theory, concepts and paradigms**.

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1 Introduction

Visualizations take on an ethical dimension when they mediate consequential relationships between designers, viewers, and the people or phenomena represented in data. Visualization research and pedagogy recognize ethical data communication as a central concern. However, ethical visualization is often reduced to avoiding a set of misleading design practices—e.g., truncating the y-axis [9, 30], violating user expectations based on graphical conventions [22, 51, 55], or distorting the correspondence between data and visual encoding channels [10, 25, 33]. Coursework around “black-hat” visualizations [10] often asks students to either find or create examples of visualizations that employ such deceptive practices as an exercise to prompt reflection. Although these studies and teaching exercises offer guidance on how visualization design shapes audience interpretation, they under-examine the role of social context, the data itself, and consequences downstream from data interpretation in evaluating the moral status of a given visualization. *We posit that these omissions stem from the difficulty of simulating the real-world conditions that give visualization an ethical dimension in the controlled settings where we research and teach.*

In practice, visualization designers encounter morally gray scenarios where tensions arise between their responsibilities to different parties. For example, a data broker providing information to an environmental analyst may need to communicate peaks in the concentration of pollution while masking proprietary sensor locations (Fig. 1). The data broker is not a malicious actor seeking to deceive, but this scenario nonetheless presents them with a complex design problem where wrong the choice of visualization design breaks or risks breaking their design constraints, which reflect their responsibilities to others. For instance, the strip plot in Figure 1 fails to communicate the location of peaks due to overplotting, and both the strip plot and the density plot may reveal sensitive information about sensor locations by showing gaps in the distribution. We might deem either outcome where the data broker’s design constraints are broken unethical: failure to reveal patterns of air pollution may harm the public interest; disclosure of sensor locations may violate a contractual obligation. The specific combination of the broker’s design constraints and the data distribution itself make it difficult to choose a visualization that doesn’t break or risk a constraint. Thus, the data broker’s choice of visualization cannot be benign when their social context endows them with responsibility and not all design choices are equally successful at communicating.

In order to investigate and teach about these morally gray scenarios more directly, we contribute a way to approximate their formal structure through *Purrsuasion*: a cat-themed, browser-based role playing game where participants negotiate constrained data sharing through visualization authoring and peer-to-peer communication. Players are assigned to the role of either (i) *receiver*, a data seeker who makes a request for information, or (ii) *sender*, one of multiple data providers who author visualizations to share with the receiver subject to design constraints. Each round of *Purrsuasion* features a puzzle like the one in Figure 1, defined by a narrative scenario, dataset, and constraints on information sharing (see Sections 3.2 and 6.2). Following prior work, we conceptualize the core design trade-offs in these puzzles in terms of *data disclosure* [34]: one *data signal*, or informational target of a task, is the subject the audience’s information need (e.g., peaks), and another data signal is the subject of the designer’s data sharing constraint (e.g., gaps). Since the puzzles in *Purrsuasion* require students designing visualizations to balance *showing* one signal and *hiding* another (e.g., Fig. 1), we call them *show-hide puzzles*. To support scoring visualizations as solutions to these puzzles, we define a heuristic rubric to help classify disclosure adherence in terms of whether individual constraints are satisfied, risked, or broken (e.g., right side of Fig. 1). Together with these show-hide constraints, a puzzle’s narrative framing of the design scenario simulates the social responsibilities of data sharing. The setting of a multi-player game helps to reify this simulated responsibility by providing a communication structure and social expectation for collaborative sensemaking [6]. We design the puzzles

in *Purrsuasion* to place the receiver’s need in partial conflict with the senders’ constraint, so an ideal solution is elusive and most visualizations authored by the senders will entertain design compromises. At the end of a round, the receiver must choose which of the senders they trust most to sign a hypothetical data sharing agreement with. The game asks both the senders and receiver to weigh the ethical desirability of compromises around disclosure in visualization design, providing students an opportunity for experiential learning while also giving researchers observability into otherwise hidden aspects of visualization practice.

We view *Purrsuasion* as an instance of a broader class of **data disclosure games**: role playing games that ask participants to navigate design puzzles, each featuring a set of disclosure constraints, a provided dataset, and a narrative design scenario. In Section 6, we discuss the generalization of the game and its broad applicability as a platform for visualization teaching and research. For instance, formal puzzles defined by disclosure constraints can approximate the structure of other scenarios that don’t entail goals about privacy or sensitive information—e.g., design trade-offs in science communication, journalism, and data exploration often emerge from needing to express multiple signals at once. To support the adoption of data disclosure games, we contribute an open-source implementation of the *Purrsuasion* platform, including onboarding materials and instructor-facing tools for running, scoring, and adapting the game.

We also contribute a mixed-methods study based on our initial deployment of *Purrsuasion* in a visualization course for data science undergraduates at [institution redacted]. 27/33 students participating in the game consent to donate their interaction log data resulting from gameplay and fill out a survey reflecting on the ethics of data disclosure. In addition to characterizing students’ communication dynamics, we attend in particular to how they ideate and conceptualize solutions to our show-hide puzzles, drawing on Mehta et al.’s *taxonomy of disclosure tactics* [34] as a deductive framework for analyzing visualization design choices that impact data disclosure. Our analysis shows that students face a “*gulf of envisioning*” [48] in *visualization design*, a fundamental difficulty conceiving of an optimal solution to satisfy design goals such as disclosure constraints. We describe how this gulf precipitates design fixation in visualization authoring for senders, difficulties attributing authorial intent for receivers, and pitfalls in negotiated data sharing as a result. In evaluating student solutions and observing how students appraise each other’s visualization design choices, we find that judgments of disclosure adherence can seldom be reduced to constraint checking, but rather, the gravity of miscommunication risk must be assessed in its social context (cf. contextual integrity [36]). Our findings suggest that ethical data disclosure through visualization requires more interactive support for activities like design ideation, interpreting solutions in terms of how they may redact the underlying dataset, formulating requests for information more precisely, and trust repair in instances of miscommunication.

2 Background

We contextualize *Purrsuasion* by highlighting how it offers a new perspective on ethical visualization and visualization literacy, and how it extends a long standing tradition of educational games in HCI and related fields.

2.1 Ethical Visualization and Visualization Literacy

A growing body of research emphasizes that ethics is inseparable from a visualization’s communicative power and its consequences for how people reason, decide, and form opinions [8]. While visualization ethics spans affect, values, and politics in addition to accuracy and responsibility [14, 15, 27, 42], much of visualization research and pedagogy centers on design decisions that produce misleading visualizations that can harm or deceive viewers [9, 10, 29, 30, 33]. While learning to recognize and avoid deceptive design is essential, narrowing ethics to a focus on deception alone underrepresents the kinds of trade-offs that arise in real-world data communication, especially in *two ways*. First, a

deception-centered framing can overstate the prevalence of intentionally malicious designers—e.g., relative to a student’s future professional experiences, which we argue might entail responsibilities to avoid full transparency about sensitive data sources. We aim to understand what kinds of design practices visualization students use and consider permissible in these design scenarios. To do so, we draw on Mehta et al.’s *taxonomy of disclosure tactics* [34] to categorize the design operations that directly influence what data signals are exposed by a chart. Our study presents the first use of this framework in an empirical investigation to evaluate designers’ and audiences’ ethical reasoning about data disclosure through visualization.

Second, the ethical significance of design is fundamentally relational. Yet, research that centers both sides of the designer-audience dyad remains limited, particularly work that studies how designers and audiences jointly navigate ethical tensions in an interactive setting [41, 56]. For instance, our ways of measuring and teaching visualization literacy often incorporate ethics through a deception-focused, individual-level lens [7]. Common visualization literacy assessments such as VLAT [26], Mini VLAT [38], and CALVI [18] assess whether a viewer can extract information from common chart types, treating visualization as an isolated object to be decoded. In this framing, the potential for deception is often reduced to an individual’s ability to spot misleading graphics and critique a finished artifact. Other works such as AVEC [19] shift toward visualization construction, but still evaluate individuals in isolation rather than the designer-audience relationship. *Purrsuasion* makes a dyadic view of ethics and literacy observable by placing students in designer and audience roles where they must negotiate what counts as sufficient, persuasive, and compliant visual evidence under disclosure constraints.

2.2 Educational Games as Pedagogy and Probe

HCI has a long-standing tradition of using educational games and simulations to make complex concepts both teachable and learnable [5, 32, 54]. Building on this tradition, communication games [16, 43] offer HCI researchers a way to study how people reason under structured roles, partial information, and competing goals. Visualization research therefore stands to benefit from studying communication games: interactions in which participants make strategic choices about how and what to communicate through visualizations. Game-based approaches have already been used productively in visualization contexts [1, 2]. We draw on this tradition by presenting a game inspired in part by negotiation activities common in business school programs (e.g., [49]). Our focus in *Purrsuasion* is on how students decide to present data signals required for common analysis tasks [4, 11, 23, 44, 45, 52] in scenarios where those design choices have moral weight. By developing and deploying the game, we contribute both empirical evidence about visualization-mediated communication and a classroom-ready activity that surfaces ethical dimensions of visualization design.

3 *Purrsuasion*: A Data Disclosure Game

Purrsuasion is an instantiation of a data disclosure game that we deployed in an undergraduate visualization course. It models ethical dilemmas that arise in data-sharing contexts where a data seeker must assess a dataset’s usefulness while the data provider is constrained in what they can disclose (see Section 1). These dynamics arise when information cannot be revealed upfront because of privacy, proprietary, contractual, or policy restrictions. *Purrsuasion* explores this kind of dilemma through a simplified data marketplace that situates participants’ design judgments within a social context of data sharing. We chose the data marketplace framing because negotiating a data-sharing agreement provides a familiar and minimal structure for explaining the game mechanics. While we describe disclosure games more generally in Section 6, below we introduce the main components of *Purrsuasion*: how to play, our software implementation, disclosure puzzles in *Purrsuasion*, and our approach to evaluating solutions.

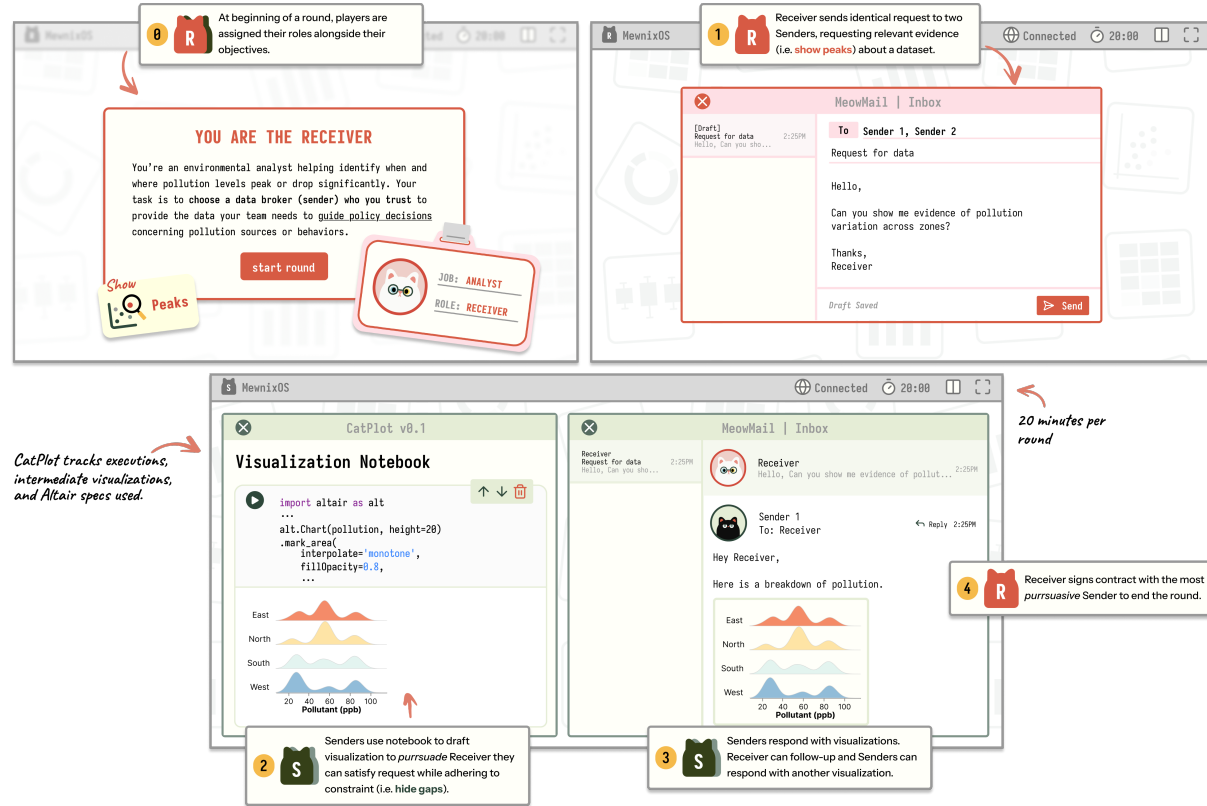


Fig. 2. An example of one round of *Purrsuasion*. The **Receiver** starts the round by sending identical queries to two **Senders**. Both **Senders** design and share visualizations that help answer the **Receiver's** query while adhering to their disclosure constraint. The **Receiver** can follow up for additional information or clarification. Finally, the **Receiver** chooses one **Sender** to award the contract and end the round.

3.1 How to Play

Purrsuasion is played in groups of three and consists of two roles: data providers that we dub **senders** and data seekers that we dub **receivers**. The game runs for three rounds, and players rotate roles across rounds. Each round has two senders and one receiver, and varies the scenario, dataset, information need, and disclosure constraint to create a different social context and ethical dilemma. Within a round, the receiver seeks visual evidence that meets an *information need* (e.g., identifying peaks in pollution data in Fig. 1), while both senders work from the same dataset to produce a visualization that meets that need without breaking a *disclosure constraint* (e.g., hiding gaps to protect proprietary collection patterns in Fig. 1). After reviewing the two responses, the receiver awards a contract to one sender and gains access to the full dataset.

Each round follows the communication sequence shown in Figure 2. Receivers open by sending identical queries to both senders, and each sender responds with a visualization that answers the query while adhering to the disclosure constraint. Receivers then ask each sender a follow-up question separately, requesting clarification or more detail, and senders reply with new or updated visualizations. Although the game can support more exchanges, our initial implementation limits each sender-receiver interaction to two responses, giving senders enough time for design ideation. The round concludes when the receiver awards the contract to one sender, along with a short written rationale for the choice. Each round of the

Table 1. Types of show-hide puzzles and their corresponding receiver-sender instructions used in the study.

Puzzle Type	Receiver Prompt	Sender Prompt
Show High Saturation, Hide Specific Locations	You're an analyst working with a tenants' rights group. Your job is to explore how retail stores are spread across the U.S. to understand where saturation is high or low. Your task is to choose a data broker (sender) who you trust to provide the data needed to understand these patterns.	You're a data broker who maintains a detailed dataset of retail stores, including their locations and counts across different areas. Your job is to share visualizations showing the dataset contains what your client (the receiver) needs. However, you should not show very fine-grained details (e.g., individual blocks or exact storefront locations) that could enable landlords or large chains to identify stores and raise rents in specific areas.
Show Peaks, Hide Gaps	You're an environmental analyst helping identify when and where pollution levels peak or drop significantly. Your task is to choose a data broker (sender) who you trust to provide the data your team needs to guide policy decisions concerning pollution sources or behaviors.	You're a data broker selling a high-resolution air-quality dataset. You want to build trust by demonstrating that the dataset contains the information your potential client (the receiver) needs. But to protect proprietary collection patterns, you cannot expose the exact location of gaps in the data distribution in your visualizations.
Show Outliers, Hide Individual Points	You're a city logistics planner allocating inspection teams for next month. Outliers in either workload or lateness could signal bottlenecks or failing warehouses. You need to identify which warehouses are atypical so you can plan targeted audits. Your task is to choose a data broker (sender) who you trust to provide this information in full upon signing a contract.	You're a data broker selling a dataset on warehouse performance. You want to build trust by demonstrating that the dataset contains the information your potential client (the receiver) needs. However, to protect supplier relationships and avoid pinpointing individual warehouses, you should hide warehouse and zone identities in your visualizations.

game presents a different disclosure puzzle, where a puzzle defines the *data* and *target signals* situated within a specific data-sharing *scenario*. See Section 6.2 for details about our puzzle design process and guidance on creating new puzzles.

Implementation. We built *Purrsuasion* as a browser-based game, lightly cat-themed (See Figure 2) to support student buy-in, in which participants log into a fictional operating system, *Mewnix OS*, that provides each role its own tools. Senders author visualizations in *Cat Plot*, an interactive notebook pre-loaded with the relevant data for the round as well as the Altair, Pandas, and NumPy libraries. Both roles exchange messages and charts through *Meow Mail*, an email-style client with convenience features that allow students to share visualizations created in *Cat Plot* with ease. Receivers award the contract through *Whisker Sign*, a document signing application that facilitates the signing of the "contract" that ends the round, and elicits receiver justification for their choice of sender. We intend that these components can be substituted in future instantiations of data disclosure games, e.g., swapping *Cat Plot*'s notebook environment for other authoring interfaces like chart pickers or AI-assisted prototypes, to support different participant populations or research goals. Our implementation houses *Purrsuasion* within a broader platform for disclosure games where instructors can manage and host their own instances of *Purrsuasion* (or future disclosure games) for their students to play. Instructor-facing features include an administrative dashboard where anonymized gameplay logs can be exported and an interface for working with the heuristic rubric described in Section 3.3. We open-source the platform such that others may contribute or collaborate with us to develop future disclosure games that enable other avenues (See Section 6) for both pedagogy and research (Source code).

3.2 Disclosure Puzzles in *Purrsuasion*

Below, we describe each show-hide puzzle we included in our initial run of *Purrsuasion* and provide examples of successful solutions vs. those that would clearly violate the disclosure constraint. Table 1 describes the puzzles along with the corresponding sender and receiver instructions.

1. Show High Saturation, Hide Specific Locations: The sender must show where store saturation is high without revealing the exact locations of individual stores. This puzzle illustrates the modifiable areal unit problem (MAUP): apparent “hot spots” can shift when the same data are aggregated into different spatial units (counties vs states vs. regions), making the consequences of choosing a particular level of aggregation directly observable. This puzzle allows us to study how senders reason about scale, justify design choices, and communicate the limits of the resulting inference. A strong solution compares multiple aggregation levels—e.g., presented in small multiples or a sequence of messages—to show which patterns persist and which are artifacts of the chosen areal unit. A clear constraint violation plots stores as points or otherwise exposes precise locations (Fig. 4A).

2. Show Peaks, Hide Gaps: The sender must communicate where pollution levels peak while concealing where measurements are missing, as gaps can reveal proprietary data collection patterns (Fig. 1). The design tension is that many natural ways to show peaks such as scatterplots, histograms, and densities can make negative space legible, effectively disclosing gaps. We found this to be a strong puzzle because it forces participants to treat visualization parameters (e.g., bin size, density bandwidth) as disclosure controls, and it lets us observe how students reason about disclosure of missingness. Good solutions emphasize peaks without preserving sample granularity—e.g., highlighting the min and max values (Fig. 4F). Constraint violations include plotting the pollution levels as points (Fig. 4D) or a histogram with small bins.

3. Show Outliers, Hide Individual Points: The sender must communicate about atypical workload or lateness while withholding the identities of warehouses and zones. This is a strong puzzle because the two data signals overlap: the extremity of a data point is often what makes it identifiable. The boundary between constraint adherence and violation is intentionally ambiguous in this puzzle, enabling us to observe how senders decide what degree of specificity is necessary for the receiver to recognize atypical warehouses, and at what point that specificity begins to reveal which warehouses these are in a way that students deem unacceptable in the scenario. Receivers must similarly reason about identifiability, privacy, and acceptable disclosure through visualizations. A good solution summarizes distributions and highlights extremes without plotting raw data—e.g., a histogram that reveal tails while preventing re-identification. A clear constraint violation plots each warehouse or zone as a point.

We designed these puzzles to probe how students reason about the ethics of data disclosure and how this is reflected in their visualization authoring and communication behavior. However, we posit that the idea of a data disclosure puzzle is a more broadly useful tool for research and teaching. We discuss generalizations of our approach beyond *Purrsuasion* itself in Section 6, including initial steps we have taken to facilitate future extensions and adoption of data disclosure games.

3.3 Evaluating Solutions

We need a general framework for evaluation that can accommodate the wide variety of visualization designs that senders may author to “purrsuade” the receiver. Our approach focuses on providing a heuristic rubric for evaluating whether the design constraints given by the show-hide objectives in *Purrsuasion* are likely satisfied, broken, or risked. This provides a foundation for thinking about whether these violations are acceptable in the social context of a given scenario. In our rubric, a visualization can receive one of three scores for each data signal. A constraint is *satisfied* when the visualization

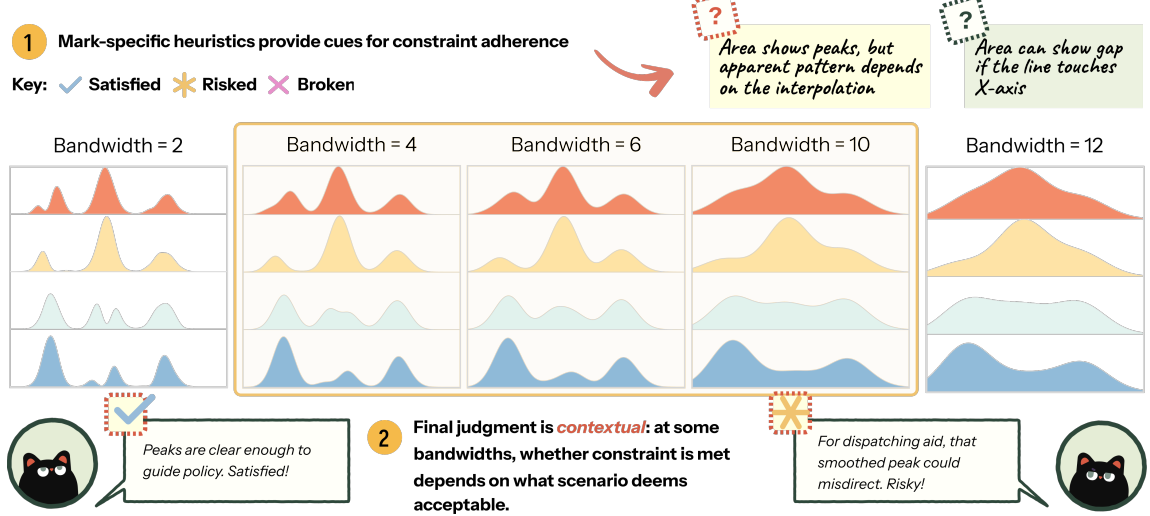


Fig. 3. Our heuristic rubric provides criteria for determining data signal presence from a visualization, but leaves the final judgment up to the grader to decide based on what the disclosure puzzle’s scenario deems acceptable. Note that the gap heuristic is not evaluated in this figure.

shows a signal meant to be shown or conceals a signal meant to be hidden. Conversely, a constraint is *broken* when the visualization conceals a signal meant to be shown or exposes a signal meant to be hidden. A constraint may also be *risky* if it neither clearly satisfies nor clearly violates a disclosure constraint. We define these cases as *risky visualizations*, or visualizations that give a receiver room to infer a forbidden signal or misread a signal they need.

More formally, for every data signal s , our rubric has *three components*:

- (1) *relevantFields* enumerates the variables needed to instantiate that signal.
- (2) *markset* captures which mark types could, in principle, encode those fields in a way that reveals the signal.
- (3) mark-specific *heuristics* provide criteria for determining signal presence from the rendered visualization, incorporating suggested cues for interpreting visual elements and underlying data transformations.

The rubric for each signal is of the following general form (see Supplemental Material for the full rubric):

ISDATASIGNAL: It is the case that a mark $m \in \text{markset}$ encodes a variable $var \in \text{relevantFields}$ such that heuristic m indicates the visualization provides sufficient evidence to detect data signal s .

Rather than treating disclosure adherence purely as a matter of whether the target signal is present in the visualization, we treat scoring as a *sociotechnical judgment* relative to the scenario set by a specific puzzle. The same visualization may be scored as risky or satisfactory depending on how consequential it would be for the receiver to infer a forbidden signal or misinterpret a needed signal (see box below). Our rubric scaffolds the contextual aspect of this judgment by identifying where communication risks may arise, but we leave it to the grader to resolve whether a given risk is acceptable.

An example of contextual judgment. Consider the ridgeline plot solution for the *Show Peaks, Hide Gaps* puzzle in Figure 3. Depending on the bandwidth used, the plot could obscure gaps in the pollution data while also smoothing or distorting the peaks that the receiver needs to identify. Whether a specific configuration of the plot constitutes an acceptable solution depends on how much the receiver’s ability to identify peaks has been compromised relative to the puzzle’s scenario, even though on a technical level the gaps have been hidden and the peaks shown (cf. contextual integrity in the privacy literature [36, 37]). For example, suppose that instead of guiding long term policy decisions (see Table 1), the receiver needs to know the peaks in order to decide where to send breathing support units during a smog event. Here, a distorted peak could misdirect aid dispersal during an emergency, so the same visualization may not be acceptable relative to the stakes. Since no fixed bandwidth indicates where this visualization turns from satisfying the constraint to breaking it, the heuristic rubric instead directs the grader to weigh the design choices in the rendered image in light of the situated task.

Whether used by a researcher or an instructor, the heuristic rubric attempts to turn a tacit judgment about disclosure into one that can be examined and compared. For research, it is an initial, deliberate attempt to make complex constraint adherence assessable, albeit manually. We discuss our arrival to this approach and preliminary experiments in determining constraint adherence in Section 6. For teaching, the heuristics serve as a probe into how students reason about disclosure. Since each heuristic names a specific way a design choice can reveal or conceal a signal, a grader applying it can see not only whether a student met a constraint but how the student understood it, and where that understanding broke down. We return to what these judgments revealed about student thinking in Section 5.

4 In Class Deployment and Analysis Approach

Teaching and research about responsible data communication has some shared challenges. Ethical dilemmas that emerge due the need for selective data disclosure [34] are (i) hard for students to gain practice with, (ii) difficult to teach about, and (iii) tricky to investigate in a controlled setting because they are inherently situated and consequential. *Purrsuasion* provides a setting where all three challenges can be addressed in tandem. The game serves as a shared, concrete activity that aligns instructional and research goals, treating our students as knowledgeable participants with moral agency whose strategies and reflections can help us study how people reason about visualization design and negotiated data sharing subject to disclosure constraints.

4.1 Class Context, Participation, and Consent

We deployed *Purrsuasion* as an in-class activity for a data visualization course in Fall 2025. The course was a core requirement for the undergraduate data science major at [institution redacted]. To ensure students had adequate preparation, we scheduled the activity late in the quarter, week 8 of 10, so that students had gained fluency with Vega-Altair and key visualization concepts prior to participating. Specifically, they had experience working with data transformations, choosing encodings, layouts, and color scales, as well as building geospatial, uncertainty, and interactive visualizations. The use of generative AI was not prohibited throughout the quarter (although the instructor gave guidance on appropriate reliance), so the activity also proceeded without any restrictions on AI use. While the scope of our empirical study was to explore how data disclosure behaviors are shaped by grammar-of-graphics-based authoring tools like Vega-Altair,¹ our

¹Prior work suggests that such grammar of graphics implementations are ill suited to design ideation about data disclosure [34, 53], so our investigation pursues situated behavioral evidence on this issue.

generalized definition of data disclosure games (see Section 6) enables the study of design ideation and problem-solving with AI-powered authoring tools, with our platform being one way of implementing such a study in the future.

We dedicated two classes to the activity. The first class included a lecture on the taxonomy of disclosure tactics developed by Mehta et al. [34], providing students with a framework for reasoning about design alternatives and the role of data transformations. We also showed a video demonstrating the game interface in an example round played by the authors. In the second class, students played the game during the whole 80 minute session. We randomized the order of rounds across student groups and randomly assigned students to groups of three. All groups completed at least two of the three rounds. See Section 3 for additional details about the game rounds. At the end of the game, the students filled out an exit survey that asked for their reflections and feedback. All students received course credit for participating and earned extra credit based on performance. See Section 3.3 for our rubric and Figure 4 for examples of scored student solutions.

We received IRB approval to analyze students' game play data. At the end of the first lecture, students were invited to consent to share their game-play data for research. We emphasized that participation was completely voluntary (i.e., a form of data donation) and would not affect their course grades. Following IRB guidelines, teaching staff then left the classroom while a non-instructor author distributed and collected paper consent forms, ensuring that the teaching staff had no knowledge of which students chose to participate. Game play data was collected from all students for grading, but our analysis only includes those who consented (27 of 33 students enrolled). The data was anonymized by the non-instructor author, and the teaching staff accessed the data of consenting students only after final grades were submitted.

4.2 Mixed Methods Analysis

We conducted a mixed-methods analysis combining qualitative analysis of gameplay artifacts with quantitative analysis of log data captured by *Cat Plot's* activity tracking (See Section 3.1). Gameplay artifacts included sender-receiver conversations, visualizations exchanged, round outcomes, and receivers' written justifications for choosing a round winner. Log data consisted of the Python code executed in each notebook cell by senders throughout a round. These data sources offered complementary views of students' disclosure choices. We analyzed both in parallel to connect observed communication and decision-making with the low-level steps of visualization design and implementation.

For the qualitative analysis, two authors conducted open coding of each sender-receiver interaction, identifying episodes relevant to our research questions and recording analytic memos to preserve context. We used the taxonomy of disclosure tactics developed by Mehta et al. [34] as deductive codes to analyze the use of data transformations in visualization design. The following *disclosure tactics* appeared in our analysis:

- **Encoded values:** Choosing what variables get represented.
- **Aggregation:** Using a statistic (e.g., count) to summarize.
- **Banding:** Using cutpoints (e.g., quantiles) to derive an interval.
- **Classification:** Partitioning a continuous variable into bins.
- **Derived values:** Combining two or more variables into one.
- **Subsampling:** Selecting a subset of records to show.
- **Smoothing:** Interpolating density with a function (e.g., KDE).

See Mehta et al. [34] for the full taxonomy and Section 7 for reflections on why *Purrsuasion* only elicits a subset of tactics. We met regularly to calibrate interpretations and maintain consistency in qualitative coding. This iterative process yielded an inductive set of themes describing student interactions during gameplay. We used these themes to characterize

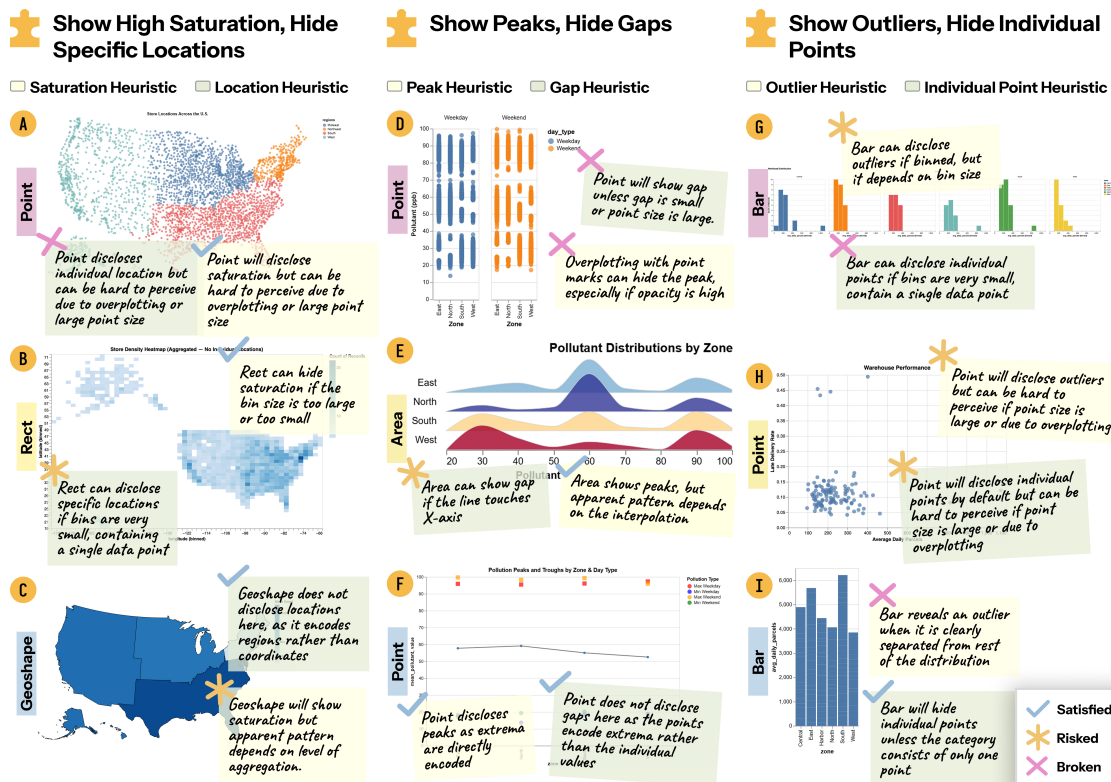


Fig. 4. Student-authored visualizations for the three puzzles in *Purrsuasion*. Instructors applied the heuristic rubric to evaluate each visualization for both the *show* signal and the *hide* signal. Each signal is evaluated to have either ✓ satisfied, ★ risked, or ✗ broken the constraint of interest.

how students negotiated disclosure, articulated information needs, and reasoned about communication, problem solving, and ethics in visualization design.

For the quantitative component, we summarized log traces to characterize senders' analysis and authoring behavior (e.g., design fixation vs. exploration) and used these summaries to corroborate and enrich the qualitative findings. See Supplemental Materials for the students' gameplay data and a codebook containing the full analysis.

5 Results

We found that students negotiating data disclosure confronted a "*gulf of envisioning*" in visualization[48]: it was difficult for students, both as senders and receivers, to conceive of an optimal solution to the show-hide puzzles we tested. This drove mirrored themes that our analysis uncovered in sender and receiver behavior, respectively. For senders, difficulty imagining what an optimal solution would look like precipitated *satisficing in visualization authoring*, where students exhibited design fixation [21, 39] and gulfs of execution around designing for disclosure with grammar of graphics APIs [34, 53]. For receivers, the same difficulty imagining an optimal solution precipitated an *inability to attribute authorial intent* on the part of senders. Given the information asymmetry between senders and receivers and the invisibility of the sender's labor in visualization authoring, receivers tended to judge the work of senders in terms of the

utility of the visualizations they sent for the task, rather than by confidence in the sender as a moral actor. Our analysis traced these themes across students' communication, problem solving, and ethical reasoning.

5.1 Communication in Sender-Receiver Triads

We analyzed student communication to characterize (1) how receivers operationalized ambiguous data needs, (2) how senders signaled competence and trustworthiness through interaction, and (3) how interface constraints like limited exchanges shaped negotiation dynamics.

Receivers differed in how they first requested data. Across 25 receiver turns, most receivers (15/25) began the conversation by asking for task-specific information about the data, often borrowing language from the instructions we provided to them. A smaller subset (7/25) started by probing the dataset more broadly—asking what patterns or variables were present—before narrowing to a task-specific follow-up. For example, receiver S_{31} asked the senders, “*Please send data on warehouses so I can see any unusual patterns,*” and then pivoted in the second message to a more explicit outlier-focused request. Only two receivers (S_{16}, S_{18}) took a different approach by requesting for a general overview of the data before asking for specific task-related information. Taken together, these openings reflected different ways receivers approached early uncertainty about how to what evidence was available in the dataset and how to allocate a limited number of queries.

Second-turn receiver requests fell into a small set of recurring request types. In the 19 rounds where receivers sent a follow-up, their second message typically served one of five purposes. Most commonly (7/19 responses), receivers requested new evidence beyond what a sender had shown. This included asking for additional variables or alternative ways to surface the target signal—e.g., “*Are there any other variables (besides zone) that can be added to find outliers...?*” (S_{31}). This extended to puzzle-specific needs for additional evidence: in the *Show High Saturation, Hide Specific Locations* puzzle, 4/8 receivers ($S_{13}, S_{20}, S_{22}, S_{26}$) asked for population context, suggesting raw counts alone were hard to interpret for saturation. 4/19 follow-ups exhibited clarifications about how to interpret a visualization. Less frequently, receivers asked for alternative visualization design choices to better support their task—e.g., “*... more granular detail... state by state... and try to make a more diverging color scale*” (S_6). Finally, some responses (2/19 responses) requested statistical information to validate that the dataset supported their inference like S_{16} , “*... show some summary statistics... so I can be sure there are outlier warehouses?*” while others (2/19 responses) like S_{20} requested for a specific visualization type, “*Can you show me if there are any outliers by zone or if there is a smaller region? Could use a boxplot.*”

Receivers almost never integrated evidence across senders. Although each receiver interacted with two competing senders who had access to the same dataset, receivers often responded to the first visualization they saw rather than waiting to compare both senders' visualizations. This produced a first-response anchoring dynamic: the second turn was typically used to clarify or extend the first visualization received, foreclosing the opportunity to triangulate evidence across sellers or request complementary views. We expected more strategic behavior, especially when both senders responded at roughly the same time. It suggested that receivers prioritized making a single view interpretable and actionable for each sender. We also observed that the puzzle structure shaped the feasibility of cross-sender integration such as in the *Show High Saturation, Hide Specific Locations* puzzle, where sender–receiver interaction was typically brief, leaving little opportunity for iteration or for senders to provide complementary views at multiple aggregation levels. Across 25 rounds, we found only one instance of an explicit cross-sender reference. In this case, after seeing sender S_{16} 's visualization, receiver S_{29} used it to formulate a more targeted request to sender S_6 , asking about a dataset variable that had not been visualized in S_6 's initial response.

Sender text functioned primarily as narration and alignment rather than reasoning. Across 66 visualizations, over half (37/66) were accompanied by a message that largely restated what the chart depicted. 13/66 were generic handoffs such as, “*Please see attached*,” (S₆). Text more rarely functioned as coordination: in 9 responses, senders explicitly checked alignment with the receiver’s intent, e.g., “*Does this work?*” (S₁₉). When senders expressed constraints on what they can show, only two responses explicitly cited the disclosure constraint. More commonly senders attributed omissions to dataset scope or interface friction such as, “*Population isn’t included*”, (S₃₂), or “*We only have weekday vs weekend*,” (S₆), making it difficult for receivers to distinguish structural limitations from strategic withholding motivated by disclosure constraints. Senders seldom provided dataset-level context. Only one sender (S₁₈) included a brief schema description. Additionally, senders rarely articulated the reasoning behind their design decisions, such as how they determined which points counted as outliers. This ambiguity made authorial intent harder to attribute and trustworthiness harder to judge, leading receivers to seek clarification.

5.2 Sender Problem Solving

Our analysis surfaced evidence of a “gulf of envisioning” [48] in senders’ visualization authoring. Students exhibited design fixation that we interpreted as a struggle to ideate optimal solutions to meet disclosure goals. Further, students underutilized data transformations (i.e., disclosure tactics [34], see Section 4.2) as an approach to eliminating the signal subject to the disclosure constraint in each puzzle.

Senders often approached disclosure through a *vagueness-first* strategy. They began with coarse or minimal evidence and increased specificity only when asked. As S₆ explained, “*I basically tried to create graphs that were as vague as possible, and when prompted for more would only provide exactly what the person asked for.*” Students achieved this primarily through the *encoded values* disclosure tactic, which they used in all 66 exchanged visualizations to limit what variables were encoded. We found this a natural strategy in response to the disclosure constraints in *Purrsuasion*, all of which designated protected fields such as zone in the *Show Outliers*, *Hide Individual Points* puzzle.

Senders overused aggregation as a disclosure tactic. 45 out of 66 visualizations used *aggregation* to collapse data values into summary statistics, with 13 of these also using *classification* to bin values before summarizing. In our log data, aggregation accounted for 73.7% of unique executions containing data transformations. As S₂₂ described, “*I focused on aggregated views like region or broader categories, basically just providing summaries, so the receiver could still see trends without being able to pinpoint exact sources.*” Students may have gravitated toward aggregation due to ease of implementation and familiarity—e.g., bridging the gulf of envisioning with a disclosure tactic whose influence on available signals they could readily anticipate. However, using aggregation sometimes yielded risky visualizations that only partially revealed the receiver’s information need. For example, in Figure 4I, aggregating *avg_daily_parcel*s by zone protected the identity of individual warehouses but distorted the receiver’s ability to detect outliers. Interestingly, 22 out of 25 round winners used aggregation even though this produced risky visualizations in 11 out of 22 cases (Fig. 6). In contrast with the overuse of aggregation, we found that other disclosure tactics were underutilized even in scenarios where they produced an optimal solution (e.g., *banding* in Fig. 4F).

Senders anchored on “safe” chart types. In their exit surveys, many senders described selecting a chart type early on in their design process and then adjusting encodings to satisfy the disclosure requirements, rather than beginning with data transformations that reduce inferential access and then choosing a chart to match. For example, S₃₃ describes, “*I first chose a type of graph (line chart, histogram, choropleth, etc.) and then carefully thought about the encodings to make sure I satisfied the requirements. If the visualization violated my constraints, then I’d start again with this process.*” *Purrsuasion*’s log data corroborated this chart-anchoring behavior broadly: for 16 out of 27 students, the first chart type

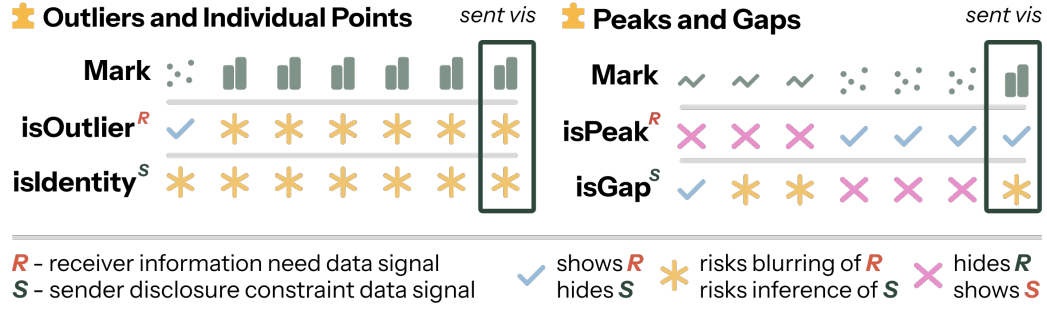


Fig. 5. From left to right, S_{33} 's design ideation sequence, leading up to the visualizations they shared.

they rendered became an anchor. Across puzzles, the anchor chart type on average accounted for 63.1% of a student's total visualizations iterations. However, the chart that students perceived as "safe" was not always so. For example, Figure 5 summarized the sequences of visualizations that S_{33} authored across two puzzles, showing that in the *Outliers and Individual Points* puzzle, they exhibited design fixation on a risky histogram. In contrast, Figure 5 showed that S_{33} struggled to ideate a solution that didn't break constraints on the *Peaks and Gaps* puzzle, suggesting that defaulting to a "safe" visualization may have reflected satisficing due to the "gulf of envisioning" [48] more so than student recognition of promising solutions.

Students exhibited uncertainty about what counted as violating a disclosure constraint. We found recurrent breakdowns in how senders translated the hiding requirement into an actionable design check. For example, in Figure 4D, plotting ppb against zone made discontinuities where data were not recorded visible to the naked eye. In other cases, similar confusion led to risky visualizations—e.g., in Figure 4H, by creating a scatterplot of avg_daily_parcel against pct_late_deliveries, the sender did not directly reveal warehouse identities but provided enough information for an interested receiver to follow up by asking about the protected signal. We found that such cases were most common in the *Outliers and Individual Points* puzzle, where the information need and disclosure constraint were least orthogonal and disclosure adherence was most open to interpretation. This uncertainty meant that students were not always aware of their performance. For example, consider S_{34} : "On the first round I had completely missed the constraint, but on the second round as sender, I made sure to check the columns I had available to me thoroughly to and eliminated columns that were unavailable due to the constraints." In reality, S_{34} broke constraints in both rounds, illustrating that students' vagueness-first strategy, which limits encoded values, was only as reliable as the sender's ability to envision solutions.

5.3 How Receivers Selected Contract Winners

Receivers based their justifications for choosing contract winners on the epistemic utility of visualizations rather than explicit ethical reasoning. For instance, 23 out of 25 receivers awarded a contract to a sender with visualizations that either satisfied or risked their information need (Fig. 6). Receivers frequently prioritized apparent interpretability. For example, S_8 preferred a visualization because, "This data more clearly conveys the information I'm looking for." Color encodings were a popular interpretability cue, with receivers saying things like, "Love the color encodings that show clearly which zone of warehouse have outliers in lateness," (S_{37}), and, "It uses the different colors for each region to visibly show spread!" (S_{24}).

Some receivers relied on judgments of whether visualized data aligned with their expectations about the domain, consistent with prior research [24]. This was illustrated by S_6 's rationale: "Although the other graph has a state-by-state stratification, I think this one is better just because the numbers make more sense. The biggest thing throwing me off of the

Hot Spots and MAUP

R: isSaturation
S: isSpecificLocation

	Mark	R	S
s36	Geo	✓	✓
s13	Geo	✓	✓
s23	Geo	✓	✓
s18	Geo+Bar	✓	✓
s10	Bar	✓	✓
s24	Bar/Arc	✓	✓
s4	Rect/Rect	✱	✓
s32	Point	✓	✱

Peaks and Gaps

R: isPeak
S: isGap

	Mark	R	S
s19	Point+Line	✓	✓
s8	Point+Line	✓	✓
s12	Line/Bar	✓	✱
s7	Line/Rect	✓	✱
s23	Box/Box	✱	✱
s16	Area/Box	✱	✱
s25	Line/Point	✱	✓
s10	Rect	✱	✓
s29	Bar	✱	✓

Outliers and Individual Points

R: isOutlier
S: isIdentity

	Mark	R	S
s37	Bar	✱	✱
s8	Point/Point	✱	✱
s33	Bar+Rect	✱	✱
s20	Point	✱	✓
s14	Bar/Point	✱	✱
s6	Point+Bar+Box/Box	✱	✱
s29	Box/Box	✓	✱
s21	Bar/Point	✱	✱

*Icons after / refer to 2nd vis sent after receiver follow-up
+ refers to concatenation of charts

✓ shows R
✱ risks blurring of R
✱ risks inference of S
✱ hides R
✱ shows S

Fig. 6. For each round winner (grouped by puzzle) we show: participant id, mark type, and disclosure adherence per signal

other map says that there are 250 stores in Alaska which absolutely makes no sense." It is important to acknowledge that the student was correctly detecting an artifact of our synthetic data generation process, as we seeded this puzzle's data using an airports dataset, producing an unusually high count in Alaska. This observation illustrates a challenge in designing realistic puzzle datasets (see Section 6.2).

Receivers valued evidence that more directly supported the task and provided sufficient data coverage. For instance, S₃₆ explained that "S₇ provided cleaner, more targeted visualizations that directly addressed the goal of identifying peaks and troughs, and adapted quickly when asked for a different view." S₂₁ emphasized, "This visualization actually provided information about late deliveries, whereas the first visualization only really gave a summary about daily parcels." Others similarly highlighted task fit, informativeness, and adaptability across senders.

5.4 Student Reasoning about Ethical Data Communication

Across exit survey reflections, students recognized the game as making ethical considerations in data communication more concrete and situational. This was often framed as revising an implicit assumption that "better" visualizations are always more detailed and granular. For example, as S₂₁ put it, "This game definitely made me realize that there are certain situations in which this would be a serious concern of privacy or ethics." Many students expressed their understanding of the ethical aspects of the game as a balancing act. For example, S₃₆ emphasized that the game "highlighted how crucial it is to communicate meaningful patterns responsibly while protecting individuals and locations from unintended exposure." Other students viewed ethics through a more conventional lens of avoiding deceptive practices. For example, S₃₄ reflected, "I think it changed my thinking because of how easy it is to try to lie and/or fabricate data to meet the needs of the receiver. You have to think carefully about the choices you make, as it is easy to make visualizations that mislead the receiver into thinking you have what they need." These responses suggest that the game achieved its educational objective of promoting nuance in the way that students think about visualization ethics without encouraging them to neglect concerns about misleading their audience.

Students also viewed the incentives of the role they were assigned as crucial to understanding the moral weight of each other's actions. For example, S₃₃ said, *"It also made me think about this as a dance, where all parties must be benefited (the broker makes the sale, the receiver gets the data it needs, and privacy is kept)."* Occasionally, during gameplay, senders directly stated their constraints as way of resolving the central dilemma presented by the need for selective disclosure—e.g., *"Can't show you exact state data but you can make out general trends from this,"* (S₁₆). The tension between sender-receiver incentives became more salient once students had the opportunity to play both roles. As S₈ described, *"As the sender, it was tough to adequately communicate what the receiver wanted while still maintaining my ethical standards. Conversely, as the receiver I ultimately went for the sender who gave me the most info—my standards of ethics (maximal disclosure) might not have lined up with their specific goals."* This suggests that playing the game brought the relational aspects of visualization ethics into focus by asking students to engage in perspective taking, and that students took their moral agency seriously in playing their assigned roles.

5.5 AI for Brainstorming

We allowed students to use GenAI tools during gameplay but did not directly observe these interactions, so in the post-game survey, we asked whether they had used an LLM for assistance and invited them to share a link to their chat session. Most students did not answer this question. One plausible explanation for this behavior is social desirability bias [28], where students avoid reporting AI use because they believe may be judged negatively.

While direct self-reports were sparse, we analyzed available responses to understand how students used AI to brainstorm and solve problems around data disclosure. Students framed LLMs as useful for generating candidate approaches under a show-hide requirement. As S₂₆ put it: *"I asked the LLM I was speaking with for advice regarding how to approach the problem given the constraint, and it provided a list of different strategies I could use (as there are many ways to solve a problem). I then thought critically about which one would make most sense for the case, and wrote code + asked the LLM for help to achieve that purpose."* This suggests that AI assistance may have entered at multiple points in the pipeline—idea generation, code drafting, and incremental debugging—rather than serving as a single "one-shot" generator. GenAI "code smell" hinted at these multiple uses as well, but we refrain speculating at length about these cues. Evaluating AI use in visualization authoring was not an objective for our study, but we recognize its importance and discuss how future work could use data disclosure games to investigate this (see Section 6).

6 Extending and Generalizing Data Disclosure Games

Whereas *Purrsuasion* focuses on using show-hide puzzles to probe ethical visualization design, we envision that the game's structure can accommodate a wide variety of teaching and research activities. For example, since we first ran the game, colleagues have approached us about adapting the game platform to teach students about statistical analysis and avoiding common pitfalls in data interpretation. To support such future use cases, we generalize the core design decisions behind *Purrsuasion* to define a broader class of **data disclosure games**: role playing games where participants attempt to navigate a series of design puzzles, each defined by a narrative scenario, dataset, and constraints on information sharing.

6.1 What Makes a Disclosure Game?

We define a disclosure game through five components that specify who participates, what kinds of information-sharing choices they must navigate, how players interact, and the particular problems through which those choices are enacted.

- (1) **Roles** differ in their access to data, instructions, or decision-making authority. Senders have access to data and communicate information through charts, while receivers evaluate whether those responses satisfy their information needs.
- (2) **Relationships** determine if players work independently, compete, or collaborate. Relationships between senders and receivers may also vary in the extent to which their goals are aligned.
- (3) **Disclosure Objective** specifies the communication goals for data signals in a visualization, such as showing one signal while hiding another or showing multiple signals simultaneously.
- (4) **Communication Structure** specifies how information moves among players, including who may communicate with whom, what forms communication may take, and how interaction unfolds over time.
- (5) **Disclosure Puzzle** is a concrete information-sharing problem through which players encounter the disclosure objective. A puzzle situates particular *data* and *target signals* within a *scenario* that gives the disclosure constraints meaning and consequences.

Below, we present two variations of data disclosure games that demonstrate the breadth of their design space and their utility for research and teaching. See Supplemental Materials for additional examples. Because the game components change in tandem to represent new design scenarios, each example varies most if not all components. In each case, we highlight the most important changes relative to *Purrsuasion*.

Forecasting for a Decision-Maker: Imagine a game with puzzles where the receiver needs to make a decision, e.g., about where to allocate limited resources, and the senders provide public forecasts with uncertainty from predictive modeling to support this choice. Such decision-making scenarios are common in policy settings such as disaster relief. The core dilemma is for the senders to author statistical graphics that preserve patterns in the data that are relevant to the receiver’s decision. Relative to *Purrsuasion*, this game’s largest changes concern the *disclosure objective* (i.e., show multiple signals), *communication structure* (i.e., forecasts are shared in a public forum), and *target signals* (i.e., forecasts with uncertainty) in the disclosure puzzles. Deploying this game would enable visualization researchers to study how students use sampling- and modeling-based disclosure tactics [34] for communicating statistical analysis and how they integrate and interpret uncertainty visualizations from multiple public sources to make decisions. Similar activities could be used to teach applied regression in data science coursework.

Medical Risk Communication: Imagine a game where the sender is a doctor using historical data to communicate with a diabetic patient (i.e., receiver) about the risks associated with different insulin delivery plans. Ideally, the sender and receiver collaborate to develop a personalized care plan—e.g., the patient asks questions, expresses concerns, and shares personal routines or preferences, thus shaping which visualizations are useful and how they should be explained. The core dilemma is for the sender and receiver to engage in mutual sensemaking. Relative to *Purrsuasion*, this game’s largest changes concern the *roles* (i.e., one-to-one information sharing), the *relationship* (i.e., both the doctor and the patient collaborate towards a common goal), and the type of *data* presented in the disclosure puzzle (i.e., doctor’s public data, patient’s private information). Deploying this game would enable visualization researchers to study how players collaboratively build trust and personalize evidence to support decision-making [47]. As an exercise in perspective-taking that invites reflection on design scenarios, games like this one that explore power dynamics present an opportunity to critique and revise problem framings in visualization [40].

6.2 Designing Disclosure Puzzles

At the core of every disclosure game is a **disclosure puzzle**: the design problem players must navigate through the game.

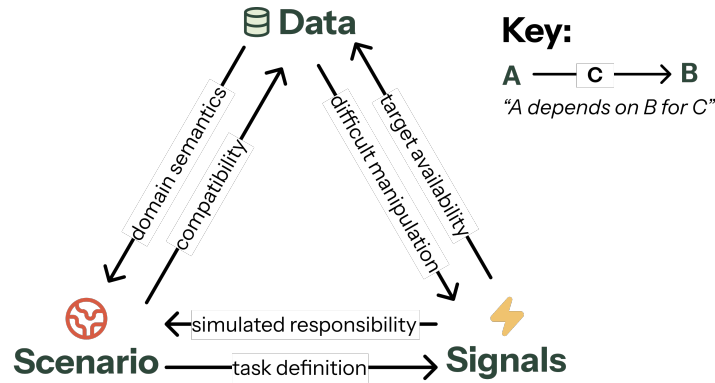


Fig. 7. Elements of disclosure puzzle are interrelated and choosing one requires carefully considering how it affects the others.

A disclosure puzzle has three elements:

- (1) **Scenario** situates the puzzle in a consequential data-sharing context. The scenario explains why one player needs the information, why another player faces limitations or obligations around sharing it, and what hypothetical consequences would follow from their choices.
- (2) **Target signals** are the facts or relationships in the data that the game is about (e.g., peaks, gaps, outliers, clusters) and slot into the disclosure objective set by the game (e.g., show-hide or show-show).
- (3) **Data** provides information that players analyze, visualize, communicate, and evaluate. The data must contain the signals implicated by the puzzle and permit multiple plausible visualization strategies.

Together, these elements lend a disclosure game a moral weight that can influence visualization design ideation. In pedagogical settings, the puzzle shapes the concepts and trade-offs that students encounter, while in research settings it shapes the behaviors and judgments that investigators are able to observe. Below we walk through how a designer might craft a disclosure puzzle, pointing out opportunities for design variation. A puzzle's scenario, target signals, and data rely on each other to create a cohesive design problem, so puzzle design is iterative rather than a strict sequence (Figure 7). We use the *Show Peaks, Hide Gaps* puzzle as a running example. We then step back to consider how the choices across these elements together calibrate a puzzle's difficulty.

Develop a Scenario. Once the designer fixes the disclosure objective for the game (i.e. show-hide for *Purrsuasion*), they situate the puzzle in a plausible scenario that defines a setting, assigns the roles their motivations, and establishes why players are obligated to respect disclosure constraints. The scenario thereby sets the stage for a set of target signals that reify the simulated responsibilities that lend the puzzle its moral weight. For example, *Show Peaks, Hide Gaps* is set in a data marketplace where the receiver is an environmental analyst and the sender is a data broker selling access to the air quality dataset. Data brokerage is a real practice with real disclosure constraints, so the tension between supplying useful information and protecting a proprietary asset arises naturally from the domain.

Design Opportunity. The scenario is where a designer can connect the puzzle to a domain that matters to them, whether their own area of expertise or one their students care about, which is what can make a puzzle a meaningful problem rather than an abstract exercise. If a fitting scenario proves difficult, generative tools can help. When developing *Purrsuasion*, we gave GPT-5 a fixed show-hide objective and asked it to propose settings where that tension might plausibly arise (see Supplemental Material).

Choose the Target Signals. The designer then chooses the data signals that instantiate the disclosure objective within the scenario (Figure 7). Target signals abstract the puzzle’s analytic tasks a visualization can support, such as finding an anomaly or characterizing a distribution [4, 45]. The disclosure objective determines which target signals are feasible as a pair. Furthermore, how sharply the two signals cross—i.e., how much the disclosure objective creates a design tension (See Figure 3)—depends on the dataset, so signal choice and dataset construction inform each other (Figure 7). For example, in *Show Peaks, Hide Gaps*, the challenge is that obscuring gaps can also distort the peaks.

Design opportunity. The signals a designer chooses determine what students reason about during design ideation. In a show-hide puzzle, pairing peaks with gaps asks them to reason about hiding missing data until a data sharing agreement is signed, while in a show-show puzzle for example, pairing a forecast with its uncertainty asks them to present two signals faithfully at once. Across disclosure objectives, signal choice is how a designer encodes underlying learning objective of the puzzle, whether it is reasoning about sensitive data or informed decision-making.

Construct the Dataset. With the target signals chosen, a designer builds the dataset that realizes them. A designer begins with a synthetic or existing dataset whose variables suit the scenario, then appends, filters, or perturbs records to bring the signals out, inspecting candidate visualizations after each change. For example, for *Show Peaks, Hide Gaps*, we simulated a small dataset of pollutant readings whose variables fit the air-quality scenario: a reading identifier, a city zone, a day type, and a concentration in parts per billion. We seeded the concentration distribution with several peaks and left gaps between them, revising until the peaks stayed legible and the gaps remained present in the data.

Design Opportunity. Building from a real data source in the scenario’s domain exposes students to the idiosyncrasies of measurement, sampling, and reporting within it. Synthetic data offers finer control over signal placement and frees a designer to construct puzzles that no available dataset would support. Datasets must be checked for plausibility, since values that strain belief draw attention to the data rather than the design choices the puzzle means to probe. Generative tools can scaffold this, though a designer should verify the result (see Supplemental Material for our data generation process.)

Calibrating Puzzle Difficulty. The three elements above jointly determine how difficult it is to solve a given puzzle. We define a puzzle’s difficulty in terms of the space of visualizations that satisfy all of its constraints. For example in *Purrsuasion*, the proportion of visualizations in a design space that can violate either the receiver’s information need, the sender’s disclosure constraint, or both acts as a proxy for difficulty. A wide space of constraint-satisfying visualizations may make the puzzle trivial, offering no design tension to reason through. Conversely, a narrow space of constraint-satisfying visualizations may make it impossible to play in good faith, potentially pushing a student toward deception. Generally, signals that sit near one another in the data domain narrow the space of possible visualizations, since a data transformation that alters one tends to alter the other. An instructor or researcher deploying a data disclosure game can tune these properties in the data to set a rough difficulty. In *Show Peaks, Hide Gaps*, we sharpened the puzzle by overlapping peaks in one zone with gaps in another zone, narrowing the constraint-satisfying subset of possible visualization designs without emptying it.

Design Opportunity. Note that the space of visualizations is understandably bound by the specific authoring tool one chooses to orchestrate the game. That is, the same signal pair can be a hard puzzle under a rich charting grammar and impossible under a fixed menu of chart types. The expressiveness an instructor grants players tunes difficulty, and is one of the choices through which an implementation can adapt a disclosure game to its own ends (see Section 3.1).

6.3 Generalizing Evaluation of Disclosure Adherence

Automating the evaluation of whether a given visualization technically shows or hides a given data signal remains a research frontier. We attempted to implement the constraint logic in our rubric as an extension of Draco [35], a system that encodes hard and soft constraints over the Altair specification. However, what a visualization discloses exceeds the chart specification [34, 53], requiring an extended version of Draco that reads all of the data processing behind a chart. Furthermore, disclosure adherence depends on interactions between a visualization’s marks and the dataset (e.g., a histogram that reveals or hides a signal depending on its bin size [11]), which requires runtime evaluation rather than prospective program analysis to judge. We deemed these challenges beyond the scope of our investigation, however, solving them would enable automated analysis of puzzle difficulty (see Section 6.2) and would enable scoring tools that help instructors and researchers reason through the technical aspect of our rubric (see Section 3.3).

7 Discussion

By developing and deploying *Purrsuasion*, we advance methodology and knowledge around visualization authoring and interpretation in scenarios that involve negotiated data disclosure. The game platform itself provides observability into how participants navigate visualization design and interpersonal correspondence around data sharing when balancing simulated tensions between responsibilities to different parties. More broadly, our framework of designing data disclosure games lays the foundation for new ways of studying visualization (see Section 6). To facilitate adoption of data disclosure games like *Purrsuasion*, we contribute an open-source, extensible game implementation with onboarding materials and guidance on game and puzzle design for instructors and researchers (see Supplemental Materials).

Based on our deployment of *Purrsuasion* in a visualization class for data science undergraduates at [institution redacted], we draw out implications of our study for trust, design ideation, query formulation, and visualization pedagogy.

Receivers face an intent attribution gap when judging visualizations. In most rounds of gameplay, receivers assess the utility of a visualization for meeting their information need more readily than they assess the intentions, constraints, or design process of the senders who created it. Accordingly the utility of a visualization serves as a proxy for the trustworthiness of the sender. Although receivers in *Purrsuasion* could in principle triangulate claims across sender responses and treat disagreement as a signal for further scrutiny, we only observe one instance of this strategy. Future work should explore ways of helping audiences interpret visualizations more defensively, e.g., by enabling them to reason proactively about (i) what a visualization might not show them by design or (ii) how difficult it would be to verify a suspected signal in a risky visualization. More broadly, visualization tools should support trust formation not only through surface cues like readability, but through analysis of the design process behind an image.

Disclosure constraints lead senders to satisfice rather than explore designs. Senders tend to settle on the first visualization they find that seems to satisfy their disclosure constraint, rather than exploring a broader range of possible solutions. While disclosure constraints narrow the available design space, they do not by themselves explain the degree of design fixation we observe. Our findings instead suggest that a central difficulty is ideational: senders struggle to

envision multiple acceptable ways of satisfying both the receiver’s information need and the disclosure constraint at once. Although this may be partially attributable to the 20-minute round structure in our deployment of *Purrsuasion*, it also suggests opportunities for future work on authoring interfaces supporting rapid ideation under disclosure constraints (cf. recent AI-assisted visualization tools [12, 31, 50]).

Receivers struggle to articulate information need. We find that receivers mostly treat written requests for information as narrow queries for relevant evidence, but lacking initial insight into available data, they seldom formulate these queries with a beneficial level of precision. Instead, receivers make vague references to the data signal named in their instructions. They rarely use their first turn to request a summary that might inform a strong follow-up question, in contrast to the “overview first” [45] opening move in most visual data exploration workflows. Future work should investigate how people formulate queries in negotiated data sharing when communication timing, interface conditions, and task framing vary. More broadly, visualization research should not assume that data seekers will spontaneously ask information-optimal questions and instead develop explicit support for query formulation.

Risky visualization is a social problem. Our findings show that negotiated data disclosure produces visualizations that cannot be classified as simply ethical or deceptive. Rather, visualizations can create data communication risks that range from benign to serious depending on sociotechnical factors such as the data context and the relationships [3] of different stakeholders. For this reason, we argue that work on visualization authoring, literacy, and pedagogy should expand from a focus on individual chart construction and interpretation toward an emphasis on the mutuality of how designers and audiences reason together given asymmetric access to data. In such contexts, sharing a risky visualization should be understood as a valid *epistemic action* that helps to coordinate knowledge between designer and audience. We argue that the resulting risks, e.g., of disclosing sensitive information or miscommunicating, are not mere failure modes but should be studied in terms of their social consequences and opportunities for repair.

7.1 Data Disclosure Games as a Boundary Object

Data disclosure games, like *Purrsuasion*, offer distinct methodological benefits for visualization research by affording an examination of shared epistemic uncertainties between students and researchers. Functioning as a boundary object [46], data disclosure games foster a connection between research, teaching, and practice. For students, it prompts perspective-taking, allowing them to acknowledge and navigate contextualized design tensions [49]. For researchers and educators, the lack of canonical solutions or a prescribed design processes fosters epistemic humility, creating opportunities to learn from students [6]. For instance, analyzing student solutions shaped our understanding of how puzzle dimensions like the scenario and dataset shape judgments of whether a risky visualization builds trust with the receiver in benign ways vs. represents an ethical violation of a simulated responsibility. Similarly, data disclosure puzzles create opportunities for students and instructors to discuss the ethics of data access ownership—e.g., our *Peaks and Gaps* puzzle surfaces the real-world friction around intellectual property claims on public interest data about air quality [13].

Viewing *Purrsuasion* as a boundary object also allowed us to explore a richer conception of visualization ethics for both research and pedagogy. It demonstrates that educational activities about visualization ethics can cultivate reasoning about morally gray scenarios without coercing students to play out ethically compromising situations. In this regard, we made choices in our puzzle design sensitive to human values and the social contexts of technology use [17]. Students act as direct stakeholders negotiating a contract, while constraints over the puzzle’s dataset function as proxies for indirect stakeholders who are vulnerable data subjects. For research, modeling and capturing these scenarios through *Purrsuasion* provides a sandbox for observing how values embedded during the design process transform or erode as users translate abstract ideals through technology [20].

7.2 Limitations

The study was conducted under several important constraints. Because we deployed the game within an 80 minute class session, we had to balance the number of puzzles against the time students had to engage meaningfully with each round. This also led us to limit the receiver to a single follow-up question, so that senders could have time to produce a reasonable visualization. As a result, the interaction patterns we observed reflect a time-bounded form of problem solving. Future work should examine how reasoning about disclosure unfolds in settings where players have more time and no fixed message limits.

By situating the game as a mock data marketplace, we centered puzzle design on concerns such as privacy and data quality. As in any empirical study, the cases we sample shape the phenomena we are able to observe. Our findings thus speak most directly to settings in which data seekers must judge whether limited disclosure provides enough confidence to proceed with a purchase that would later reveal the full dataset. Future work should apply data disclosure games to other constrained data communication settings, including cases that require statistical inference, communication of multiple data signals at once, or interpretation by viewers with varying levels of visualization literacy.

The study was conducted in an undergraduate visualization course, so the behaviors we observed reflect a particular participant population. Our participants likely occupied an intermediate position between visualization novices and practitioners, with greater technical fluency and visualization literacy than the average audience, but less experience than data scientists or visualization experts. Future work should investigate how visualization authoring and data communication behaviors vary across populations with different levels of technical expertise, visualization experience, and domain knowledge.

Finally, while we allowed students to use GenAI tools during gameplay, we did not directly observe their interactions with these systems. Self-reports of AI use were sparse, in part because reporting was voluntary and students did not consistently maintain or provide complete records of their AI conversations. Consequently, our understanding of how AI influenced problem-solving, idea generation, and coding errors was limited to these self-reports and post-hoc analysis of Altair code snippets. While not the focus of this study, future work could use our platform to treat GenAI as part of the communication structure within a data disclosure game, providing a dedicated AI interface alongside or instead of a programmatic visualization authoring environment. This would make AI-mediated reasoning, prompting, and iteration directly observable.

8 Conclusion

We contribute *Purrsuasion*, an open-source visualization game for studying how students navigate ethical data communication and negotiated data disclosure. Our findings show that students authoring visualizations often satisfied because it was difficult to envision solutions that simultaneously satisfied all design constraints. Students acting as the audience for these visualizations often struggled to infer authorial intent and instead based trust on the utility of visualizations for their task. To evaluate student solutions to puzzles presented in the game, we developed a heuristic rubric that supports sociotechnical judgments of disclosure adherence. Together, these contributions position *Purrsuasion* as both a research instrument and a pedagogical tool for studying and teaching ethical visualization under negotiated data disclosure.

References

- [1] Eytan Adar and Elsie Lee-Robbins. 2023. Roboviz: A Game-Centered Project for Information Visualization Education. *IEEE Transactions on Visualization and Computer Graphics* 29, 1 (Jan. 2023), 268–277. doi:10.1109/TVCG.2022.3209402
- [2] Patrick Adelberger, Oleg Lesota, Klaus Eckelt, Markus Schedl, and Marc Streit. 2025. Iguanodon: A Code-Breaking Game for Improving Visualization Construction Literacy. *IEEE Transactions on Visualization and Computer Graphics* 31, 9 (2025), 5713–5725. doi:10.1109/TVCG.2024.3468948

- [3] Derya Akbaba, Lauren Klein, and Miriah Meyer. 2025. Entanglements for Visualization: Changing Research Outcomes through Feminist Theory. *IEEE Transactions on Visualization and Computer Graphics* 31, 1 (Jan. 2025), 1279–1289. doi:10.1109/TVCG.2024.3456171
- [4] R. Amar, J. Eagan, and J. Stasko. 2005. Low-level components of analytic activity in information visualization. In *IEEE Symposium on Information Visualization, 2005. INFOVIS 2005*. 111–117. doi:10.1109/INFVIS.2005.1532136
- [5] Shurui Bai, Khe Foon Hew, and Biyun Huang. 2020. Does gamification improve student learning outcome? Evidence from a meta-analysis and synthesis of qualitative data in educational contexts. *Educational Research Review* 30 (2020), 100322. doi:10.1016/j.edurev.2020.100322
- [6] Eva Brandt and Jörn Messeter. 2004. Facilitating collaboration through design games. In *Proceedings of the Eighth Conference on Participatory Design: Artful Integration: Interweaving Media, Materials and Practices - Volume 1* (Toronto, Ontario, Canada) (PDC 04). Association for Computing Machinery, New York, NY, USA, 121–131. doi:10.1145/1011870.1011885
- [7] Jorge D. Camba, Pedro Company, and Vetricia L. Byrd. 2022. Identifying Deception as a Critical Component of Visualization Literacy. *IEEE Computer Graphics and Applications* 42, 1 (2022), 116–122. doi:10.1109/MCG.2021.3132004
- [8] Michael Correll. 2019. Ethical Dimensions of Visualization Research. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300418
- [9] Michael Correll, Enrico Bertini, and Steven Franconeri. 2020. Truncating the Y-Axis: Threat or Menace?. In *Proc. CHI* (Honolulu). ACM, New York, 1–12. doi:10.1145/3313831.3376222
- [10] Michael Correll and Jeffrey Heer. 2017. Black hat visualization. In *Workshop on Dealing with Cognitive Biases in Visualisations (DECISive)*, *IEEE VIS*, Vol. 1. 10. <https://idl.cs.washington.edu/files/2017-BlackHatVis-DECISive.pdf>.
- [11] Michael Correll, Mingwei Li, Gordon Kindlmann, and Carlos Scheidegger. 2019. Looks Good To Me: Visualizations As Sanity Checks. *IEEE Transactions on Visualization and Computer Graphics* 25, 1 (2019), 830–839. doi:10.1109/TVCG.2018.2864907
- [12] Victor Dibia. 2023. LIDA: A Tool for Automatic Generation of Grammar-Agnostic Visualizations and Infographics using Large Language Models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, Danushka Bollegala, Ruihong Huang, and Alan Ritter (Eds.). Association for Computational Linguistics, Toronto, Canada, 113–126. doi:10.18653/v1/2023.acl-demo.11
- [13] Sebastian Diez, Thomas J. Bannan, Miriam Chacón-Mateos, Pete M. Edwards, Valerio Ferracci, Doğuşhan Kılıç, Alastair C. Lewis, Carl Malings, Nicholas A. Martin, Olalekan Popoola, Colleen Rosales, Sean Schmitz, Philipp Schneider, and Erika von Schneidmesser. 2025. A framework for advancing independent air quality sensor measurements via transparent data generating process classification. *npj Climate and Atmospheric Science* 8 (2025), 285. doi:10.1038/s41612-025-01161-2
- [14] Marian Dörk, Patrick Feng, Christopher Collins, and Sheelagh Carpendale. 2013. Critical InfoVis: exploring the politics of visualization. In *CHI '13 Extended Abstracts on Human Factors in Computing Systems* (Paris, France) (CHI EA '13). Association for Computing Machinery, New York, NY, USA, 2189–2198. doi:10.1145/2468356.2468739
- [15] Fabian Ehmel, Viktoria Brüggemann, and Marian Dörk. 2021. Topography of violence: considerations for ethical and collaborative visualization design. In *Computer Graphics Forum*, Vol. 40. Wiley Online Library, 13–24. doi:10.1111/cgf.14285
- [16] Meta Fundamental AI Research Diplomacy Team (FAIR)†, Anton Bakhtin, Noam Brown, Emily Dinan, Gabriele Farina, Colin Flaherty, Daniel Fried, Andrew Goff, Jonathan Gray, Hengyuan Hu, Athul Paul Jacob, Mojtaba Komeili, Karthik Konath, Minae Kwon, Adam Lerer, Mike Lewis, Alexander H. Miller, Sasha Mitts, Adithya Renduchintala, Stephen Roller, Dirk Rowe, Weiyan Shi, Joe Spisak, Alexander Wei, David Wu, Hugh Zhang, and Markus Zijlstra. 2022. Human-level play in the game of <i>Diplomacy</i> by combining language models with strategic reasoning. *Science* 378, 6624 (2022), 1067–1074. arXiv:<https://www.science.org/doi/pdf/10.1126/science.ade9097> doi:10.1126/science.ade9097
- [17] Batya Friedman, Peter H. Kahn, Alan Borning, and Alina Hultgren. 2013. Value Sensitive Design and Information Systems. In *Early Engagement and New Technologies: Opening up the Laboratory*, Neelke Doorn, Daan Schuurbiers, Ibo van de Poel, and Michael E. Gorman (Eds.). Springer Netherlands, Dordrecht, 55–95. doi:10.1007/978-94-007-7844-3_4
- [18] Lily W. Ge, Yuan Cui, and Matthew Kay. 2023. CALVI: Critical Thinking Assessment for Literacy in Visualizations. In *Proc. CHI* (Hamburg, Germany). ACM, New York, Article 815, 18 pages. doi:10.1145/3544548.3581406
- [19] Lily W. Ge, Yuan Cui, and Matthew Kay. 2025. AVEC: An Assessment of Visual Encoding Ability in Visualization Construction. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (CHI '25). Association for Computing Machinery, New York, NY, USA, Article 1166, 16 pages. doi:10.1145/3706598.3713364
- [20] Sucheta Ghoshal and Sayamindu Dasgupta. 2023. Design Values in Action: Toward a Theory of Value Dilution. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference* (New York, NY, USA, 2023-07-10) (DIS '23). Association for Computing Machinery, 2347–2361. doi:10.1145/3563657.3596122
- [21] David G. Jansson and Steven M. Smith. 1991. Design fixation. *Design Studies* 12, 1 (1991), 3–11. doi:10.1016/0142-694X(91)90003-F
- [22] Sarah H. Kerns and Jeremy B. Wilmer. 2021. Two graphs walk into a bar: Readout-based measurement reveals the Bar-Tip Limit error, a common, categorical misinterpretation of mean bar graphs. *Journal of Vision* 21, 12 (11 2021), 17–17. doi:10.1167/jov.21.12.17
- [23] Younghoon Kim and Jeffrey Heer. 2018. Assessing Effects of Task and Data Distribution on the Effectiveness of Visual Encodings. *Computer Graphics Forum* 37, 3 (2018), 157–167. arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1111/cgf.13409> doi:10.1111/cgf.13409
- [24] Yea-Seul Kim, Katharina Reinecke, and Jessica Hullman. 2018. Data Through Others' Eyes: The Impact of Visualizing Others' Expectations on Visualization Interpretation. *IEEE Transactions on Visualization and Computer Graphics* 24, 1 (2018), 760–769. doi:10.1109/TVCG.2017.2745240
- [25] Gordon Kindlmann and Carlos Scheidegger. 2014. An Algebraic Process for Visualization Design. *IEEE Trans. Visual Comput. Graphics* 20, 12 (2014), 2181–2190. doi:10.1109/TVCG.2014.2346325

- [26] Sukwon Lee, Sung-Hee Kim, and Bum Chul Kwon. 2017. VLAT: Development of a Visualization Literacy Assessment Test. *IEEE Trans. Visual Comput. Graphics* 23, 1 (2017), 551–560. doi:10.1109/TVCG.2016.2598920
- [27] E. Lee-Robbins and E. Adar. 2023. Affective Learning Objectives for Communicative Visualizations. *IEEE Trans. Visual Comput. Graphics* 29, 01 (jan 2023), 1–11. doi:10.1109/TVCG.2022.3209500
- [28] Yier Ling, Alex Kale, and Alex Imas. 2026. Underreporting of AI Use: The Role of Social Desirability Bias. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. Association for Computing Machinery, New York, NY, USA, Article 62, 22 pages. doi:10.1145/3772318.3791073
- [29] Leo Yu-Ho Lo, Ayush Gupta, Kento Shigyo, Aoyu Wu, Enrico Bertini, and Huamin Qu. 2022. Misinformed by visualization: What do we learn from misinformative visualizations?. In *Comput. Graphics Forum*, Vol. 41. Wiley Online Library, 515–525. doi:10.1111/cgf.14559
- [30] Sheng Long and Matthew Kay. 2024. To Cut or Not To Cut? A Systematic Exploration of Y-Axis Truncation. In *Proc. CHI (Honolulu, HI, USA) (CHI '24)*. ACM, New York, Article 207, 12 pages. doi:10.1145/3613904.3642102
- [31] Paula Maddigan and Teo Susnjak. 2023. Chat2VIS: Generating Data Visualizations via Natural Language Using ChatGPT, Codex and GPT-3 Large Language Models. *IEEE Access* 11 (2023), 45181–45193. doi:10.1109/ACCESS.2023.3274199
- [32] Thomas W. Malone. 1981. Toward a Theory of Intrinsically Motivating Instruction. *Cognitive Science* 5, 4 (1981), 333–369. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1207/s15516709cog0504_2 doi:10.1207/s15516709cog0504_2
- [33] Andrew McNutt, Gordon Kindlmann, and Michael Correll. 2020. Surfacing Visualization Mirages. In *In Proc. CHI (, Honolulu)*. ACM, New York, 1–16. doi:10.1145/3313831.3376420
- [34] Krishna Mehta, Gordon Kindlmann, and Alex Kale. 2026. Designing for Disclosure in Data Visualizations. *IEEE Transactions on Visualization and Computer Graphics* 32, 1 (2026), 1317–1327. doi:10.1109/TVCG.2025.3634781
- [35] Dominik Moritz, Chenglong Wang, Greg L. Nelson, Halden Lin, Adam M. Smith, Bill Howe, and Jeffrey Heer. 2019. Formalizing Visualization Design Knowledge as Constraints: Actionable and Extensible Models in Draco. *IEEE Transactions on Visualization and Computer Graphics* 25, 1 (2019), 438–448. doi:10.1109/TVCG.2018.2865240
- [36] Helen Nissenbaum. 2009. *Privacy in Context: Technology, Policy, and the Integrity of Social Life*. Stanford University Press, USA.
- [37] Helen Nissenbaum. 2019. Contextual Integrity Up and Down the Data Food Chain. *Theoretical Inquiries in Law* 20, 1 (2019), 221–256. doi:doi:10.1515/til-2019-0008
- [38] Saugat Pandey and Alvitta Ottley. 2023. Mini-VLAT: A Short and Effective Measure of Visualization Literacy. *Computer Graphics Forum* 42, 3 (2023), 1–11. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1111/cgf.14809 doi:10.1111/cgf.14809
- [39] Paul Parsons, Prakash Shukla, and Chorong Park. 2021. Fixation and Creativity in Data Visualization Design: Experiences and Perspectives of Practitioners. In *2021 IEEE Visualization Conference (VIS)*. 76–80. doi:10.1109/VIS49827.2021.9623297
- [40] Paul C. Parsons and Prakash Chandra Shukla. 2026. Beyond Problem Solving: Framing and Problem-Solution Co-Evolution in Data Visualization Design. *IEEE Transactions on Visualization and Computer Graphics* 32, 1 (2026), 24–34. doi:10.1109/TVCG.2025.3633866
- [41] Nanayakkara Priyanka, Bater Johes, He Xi, Hullman Jessica, and Rogers Jennie. 2022. Visualizing Privacy-Utility Trade-Offs in Differentially Private Data Releases. *Proceedings on Privacy Enhancing Technologies* 2022, 2 (03 2022), 601–618. doi:10.2478/popets-2022-0058
- [42] Nathalie Henry Riche, Christophe Hurter, Nicholas Diakopoulos, and Sheelagh Carpendale. 2018. *Data-driven storytelling*. CRC Press. doi:10.1201/9781315281575
- [43] Jens Riegelsberger, M. Angela Sasse, and John D. McCarthy. 2003. The researcher's dilemma: evaluating trust in computer-mediated communication. *International Journal of Human-Computer Studies* 58, 6 (2003), 759–781. Trust and Technology. doi:10.1016/S1071-5819(03)00042-9
- [44] Alper Sarikaya and Michael Gleicher. 2018. Scatterplots: Tasks, Data, and Designs. *IEEE Transactions on Visualization and Computer Graphics* 24, 1 (2018), 402–412. doi:10.1109/TVCG.2017.2744184
- [45] Ben Shneiderman. 2003. The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations. In *The Craft of Information Visualization*. Morgan Kaufmann, San Francisco, 364–371. doi:10.1016/B978-155860915-0/50046-9
- [46] Susan Leigh Star and James R. Griesemer. 1989. Institutional Ecology, 'Translations' and Boundary Objects: Amateurs and Professionals in Berkeley's Museum of Vertebrate Zoology, 1907–39. *Social Studies of Science* 19, 3 (1989), 387–420. arXiv:https://doi.org/10.1177/030631289019003001 doi:10.1177/030631289019003001
- [47] Hariharan Subramonyam, Bongshin Lee, Sile O'Modhrain, and Eytan Adar. 2017. Data Dialog: Facilitating Collaborative Decision Making through Data-Driven Conversations. In *Proceedings of the 11th EAI International Conference on Pervasive Computing Technologies for Healthcare*. ACM, Barcelona Spain, 440–443. doi:10.1145/3154862.3154921
- [48] Hari Subramonyam, Roy Pea, Christopher Pondoc, Maneesh Agrawala, and Colleen Seifert. 2024. Bridging the Gulf of Envisioning: Cognitive Challenges in Prompt Based Interactions with LLMs. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (Honolulu, HI, USA) (CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 1039, 19 pages. doi:10.1145/3613904.3642754
- [49] Carmen Tanner, David Schmocker, Johannes Katsarov, and Markus Christen. 2022. Educating moral sensitivity in business: An experimental study to evaluate the effectiveness of a serious moral game. *Computers & Education* 178 (2022), 104381. doi:10.1016/j.compedu.2021.104381
- [50] Yuan Tian, Weiwei Cui, Dazhen Deng, Xinjing Yi, Yurun Yang, Haidong Zhang, and Yingcai Wu. 2025. ChartGPT: Leveraging LLMs to Generate Charts From Abstract Natural Language. *IEEE Transactions on Visualization and Computer Graphics* 31, 3 (2025), 1731–1745. doi:10.1109/TVCG.2024.3368621

- [51] Edward Tufte. 1983. *The Visual Display of Quantitative Information*. Graphics Press, Cheshire, Conn. <https://www.edwardtufte.com/book/the-visual-display-of-quantitative-information/>.
- [52] Leland Wilkinson and Graham Wills. 2008. Scagnostics Distributions. *Journal of Computational and Graphical Statistics* 17, 2 (2008), 473–491. arXiv:<https://doi.org/10.1198/106186008X320465> doi:10.1198/106186008X320465
- [53] Eugene Wu and Remco Cheng. 2024. Design-Specific Transformations in Visualization. *IEEE VIS BELIV Workshop* (2024). doi:10.48550/arXiv.2407.06404
- [54] Jingzhou Ye, Yao Li, Wenting Zou, and Xueqiang Wang. 2025. From Awareness to Action: The Effects of Experiential Learning on Educating Users about Dark Patterns. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 224, 22 pages. doi:10.1145/3706598.3713493
- [55] J Zacks and B Tversky. 1999. Bars and lines: a study of graphic communication. *Memory & cognition* 27, 6 (1999), 1073–1079. doi:10.3758/BF03201236
- [56] Dongping Zhang, Jason Hartline, and Jessica Hullman. 2024. Designing Shared Information Displays for Agents of Varying Strategic Sophistication. *Proc. ACM Hum.-Comput. Interact.* 8, CSCW1, Article 42 (April 2024), 34 pages. doi:10.1145/3637319

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